Repurposing candidates for BUB1B-related mosaic variegated aneuploidy: a falsification-first decision rule with one named conditional candidate

Track 2 submission. Submissions are hypotheses for follow-up, not evidence that a medicine works. This report characterizes the variant mechanism and names one approved medicine as a conditional research hypothesis that a single decisive experiment can confirm or overturn.

> **If you are reading this as a parent, a clinician, or a patient advocate rather than as a mitosis specialist, start with `PLAIN_LANGUAGE_SUMMARY.md`.** It is one page, it carries the same conclusion, and it states what this is *not* as prominently as what it is. The predictable harm of a repurposing hypothesis that reaches a family is that it is read as a recommendation, and a 60-page expert document is not a defence against that. For what a care team should know independent of our candidate — including the published cancer-surveillance consensus that names this condition — go to `SAFETY_ANNEX.md`.

Fill slots written `{{FILL:...}}` carry patient-specific facts and are completed locally from the Phase-1 fact register; no patient-specific fact in this report is inferred by an AI system. Bold **F-** identifiers cite rows of that register (`fact_register.csv`), where each fact carries its status, basis, source, evidence strength, what would resolve it, and whether overturning it would change the therapeutic direction. The candidate denominator, the primary-hypothesis sentence, the absence of unfilled slots, and every count this report states about its own registers (claims, global-negatives, registered facts, product rows, self-tests) are recomputed from source and enforced by `make_manifest.py` before submission (§12).

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1. Executive decision

The deliverable is a **decision rule**, and under that rule a **single named conditional candidate**.

> **PRIMARY:** A repurposed medicine for BUB1B-related mosaic variegated aneuploidy earns its next experiment only if it acts in the compensating direction on the bottleneck the decisive experiment is built to measure — residual BubR1 abundance and checkpoint function, both of which are UNKNOWN in this child — passes this patient's constraint gates with every safety axis those gates cannot see declared rather than counted as clear, and rests on an approved product with a dated label. Under that rule the single named candidate is niacin (nicotinic acid), an approved NAD+ precursor acting through the SIRT2–BubR1 stabilization axis, carried as a conditional research hypothesis and not a recommendation: its oncology axis is contradictory and its growth axis unassessed. It is overturned if NAD+ repletion in patient-derived cells fails to raise BubR1 protein, fails to correct the checkpoint defect, or raises the aneuploid fraction of the surviving population — a fraction, never raw survival, because a real checkpoint rescue can raise overall survival.

Why a rule and not a list: the task is to identify an *approved medicine that is a plausible candidate for further investigation*. A list of plausible-sounding drugs does not tell a reviewer, or the child's care team, **why** one earns an experiment and another does not, nor what result would retire it. The rule does both, and it outlives this single case — it transfers to the next MVA child and to the other MVA genes (§11). The named candidate makes the rule concrete and gives reviewers a real object to scrutinize; it is deliberately carried as *conditional*, because the honest state of the evidence is that the strongest biology on this axis has never been tested with this drug, in this disease, in cells of this allele class.

What would make us wrong is stated up front (§1 component 3, and §9): this is a hypothesis built to be falsified by one experiment, not defended.

Three readers, three different things to take from this. A **researcher** should read §3 and §9: the mechanism chain, every link carrying its own evidence strength, model system and allele-class match, and the counter-evidence that could retire it — including a second SIRT2 site on BubR1 that acts in the *opposite* direction to the one this hypothesis depends on. A **clinician** should read `SAFETY_ANNEX.md` and §6: which drug classes behave differently when the

spindle checkpoint is already weak, pointed first at the published cancer-surveillance consensus that names MVA, plus one *ex vivo* experiment specified end to end. A **family member or patient advocate** should read §6 and §10: the measurement we propose has value **whatever happens to the drug** — it would be the first functional characterisation of this child's residual BubR1, readable against a published mouse dose–phenotype series. We are deliberate about how far that goes: it is a **research-grade functional characterisation, not a validated diagnostic or prognostic test**, because no human calibration exists that would turn a number from one child's cells into a prognosis. And §10 states precisely where the approved-drug space is empty, so the next team does not repeat our search. If the named candidate fails, those three things survive it. That is deliberate.

What is new here is not the engine. A gate engine that reads a case file is ordinary software, and we do not claim otherwise (§11). What we have not seen done is building a submission *designed to catch itself*. The claim ledger is generated from its sources, not typed. The release gate recomputes every count this report states about its own registers and refuses to publish if one has drifted — it has caught us. The engine reports the safety axes its own gates structurally **cannot** see, so a gate PASS can never be read as "no risk found". A citation whose full text we could not obtain is graded `asserted`, quoted for nothing, and flagged in §13. The engine is run against itself on five axes — ablation, patient perturbation, name blinding, synthetic probes and leave-one-source-out (§8b) — and two of those are assertions that fail the build, so the check cannot be skipped on a bad day; it is what turned up the most uncomfortable fact in this report, that the mechanism chain has two single points of failure. We demoted our own previous lead candidate on mechanism, and said so in every deliverable. Each of those is a place where we made it *easier*, not harder, for a reviewer to find us wrong — and that, rather than the code, is the method we are offering.

2. Patient basis — what is confirmed, assumed, and unknown

Everything downstream is conditioned on this three-layer table, so that no assumption is silently used as if it were measured. Values are filled locally; see `fact_register.csv` — 18 registered facts: 7 CONFIRMED, 2 ASSUMED, 9 UNKNOWN, each with its basis, source, evidence strength, resolution path and whether overturning it would change the therapeutic direction.

- **CONFIRMED** (measured, with source): the two *BUB1B* variants locked by our Track 1 submission, which scored a full match at rank 1 on the challenge leaderboard — a truncating allele `c.2210T>G, p.Leu737Ter (exon 17/23)` (**F-VAR-01**) and a kinase-domain missense allele `c.3006T>G, p.Asn1002Lys (exon 23/23)` (**F-VAR-02**), each on transcript `ENST00000287598.11 (MANE Select NM_001211.6)` (accession.version), genome build GRCh38. Gene: *BUB1B* (BubR1). ClinVar status of each allele: **neither variant has a ClinVar record**. Both were checked by exact-coordinate overlay against the ClinVar release dated **2026-10-04** during the VEP 116 annotation run; neither returned CLNSIG, CLNREVSTAT or CLNDN. Bounded reading: absent from that release at that date, which is not the same as absent from ClinVar today and is certainly not evidence of benignity. Two consequences matter. First, the rank-1 full match our Track 1 submission achieved was **not** a database echo — there was no record to look up, so the identification rests on the pipeline's own reasoning. Second, the pathogenicity argument in this report cannot lean on curated classification at all; it rests on the genotype matching the canonical MVA1 configuration, on the two-class model of MVA mutations (link 3), and on the predictors below — which is why every one of those is labelled with its own evidence strength rather than pooled into a verdict.
- **ASSUMED** (stated, not independently measured here) **and PREDICTED (registered UNKNOWN)**. Only two facts are ASSUMED in the register: biallelic *trans* configuration `not established by the data: short reads could not link the two variants, and parental samples are not provided; *trans* is assumed from the canonical MVA1 genotype` (**F-PHASE-01**) and recessive inheritance (**F-INH-01**); parental data are not provided for this case, so the compound-heterozygous assumption follows the canonical MVA1 genotype and is flagged as such. The remaining items in this bullet are **rule-based or computational calls whose register status is UNKNOWN**, because the quantity underneath has not been measured. We group them here because they are used the same way — never as if measured — but the register is the authority on their status. Nonsense-mediated decay of the truncating allele `NMD_predicted — the premature stop sits in exon 17 of 23, 748 nt upstream of the last exon junction, and triggers no escape rule` (**F-TX-01**, **F-TX-02**) (a rule-based call from transcript position, not an RNA measurement); missense pathogenicity predictors for the kinase-domain allele — SIFT `deleterious(0.01)` and PolyPhen `probably_damaging(0.997)`, from Ensembl VEP 116 run with `--everything` (**F-PROT-02**). These are *predicted* and are never used downstream as if measured. **REVEL and**

AlphaMissense are not reported here because they were never computed: the annotation run used VEP's built-in set plus a custom ClinVar overlay, with no dbNSFP or AlphaMissense plugin. We state the predictors we actually have rather than the ones a reader might expect. The truncating allele carries no missense predictor, correctly (**F-PROT-01**).

- **UNKNOWN** (the quantities the decisive experiment exists to measure): residual BubR1 protein abundance in this child's cells (**F-RF-01**); K668 acetylation state; spindle-assembly-checkpoint (SAC) function readout; whether a truncated BubR1 product is expressed. These are not gaps in diligence — they are unmeasurable without the patient's cells, which this team does not access, and they are exactly what §6 proposes to measure.

Phenotype context is summarized at the category level only (`an embryonal tumour, a renal mineralisation feature, growth and perinatal features, skeletal–muscle involvement, and a parental reproductive observation`, all carried by a single register row, **F-PHEN-01**, not one row per term — the per-term detail lives in the mechanism chain's phenotype nodes, where each term is either connected to a sourced edge or listed as unexplained); the four constraint-gate drivers used by the engine are `child` (**F-CLIN-age**), `history` (**F-CLIN-malig**), `nephrocalcinosis` (**F-CLIN-renal**), `restricted` (**F-CLIN-growth**). Each gate driver is registered twice — the observed clinical feature and the current functional value — because the second is usually UNKNOWN and must not be read off the first.

3. Variant mechanism: loss-of-function, the disrupted pathway, and the downstream consequence

Direction of effect: loss-of-function (hypomorphic). There is no gain-of-function evidence; a dominant-negative contribution is possible only if the truncating allele escapes decay and produces a stable fragment (scenario S3, §8). Each link carries its evidence strength (measured / predicted / asserted).

Each link below carries its own evidence strength, the system it was measured in, and — where it matters — what cuts against it. The same chain is machine-readable in `docs/mechanism_graph.yaml` and rendered in `MECHANISM.md`, where every edge also carries its counter-evidence field.

1 — BUB1B encodes BubR1, a core component of the mitotic checkpoint complex; its acetylation state acts as a switch between APC/C inhibitor and APC/C substrate, and so sets mitotic timing. *measured (human cells)* · Choi 2009, PMID 19407811

1b — BubR1's "kinase" domain is a pseudokinase. Human BUBR1 retains the catalytic triad, but putative catalysis is *dispensable* for error-free segregation; the residues that would bind ATP in a conventional kinase instead govern the protein's conformational stability. *measured (human cells)* · Suijkerbuijk 2012, PMID 22698286

1c — That domain nonetheless has a direct mitotic function, not only a structural one. It is *required* to promote KARD phosphorylation, which recruits PP2A-B56 to the outer kinetochore; mutating or removing it reduces PP2A-B56 recruitment, attenuates checkpoint silencing, and causes chromosome-alignment errors through imbalanced Aurora B activity. *measured (human cells)* · Gama Braga 2020, PMID 33207204

2 — Biallelic BUB1B variants cause MVA1; the canonical genotype is a truncating allele plus a kinase-domain missense allele, and patient cells with total PCS/MVA are demonstrably defective in the mitotic-spindle checkpoint. *measured (human genetics)* · Hanks 2004, PMID 15475955; Matsuura 2000, PMID 10877982; Kajii 2001, PMID 11746029

3 — MVA mutations fall into two empirically separable classes: those that impose a specific defect in checkpoint activity or microtubule attachment, and those that simply lower BUBR1 abundance. In the low-abundance group the substitution "consistently occurs in or near the BUBR1 kinase domain"; read with link 1b, a substitution there destabilises the fold rather than abolishing catalysis. **This is joint 1 below — it does not license the reverse inference for an untested variant, and link 1c cuts against it.** *measured (MVA patient cell lines + functional protein replacement)* · Suijkerbuijk 2010, PMID 20516114; Suijkerbuijk 2012, PMID 22698286; Gama Braga 2020, PMID 33207204

4 — Net effect is a dose (abundance) loss-of-function (hypomorph), possibly compounded by an allele-specific quality defect: in mouse embryonic fibroblasts BubR1 H/L1002P reaches virtually the same aneuploidy rate as BubR1 H/H but shows a dramatic difference in premature chromatid separation, at BubR1 levels the paper reports as very

similar — so "allelic effects beyond protein level" contribute. **Joint 2 below: three bounds, and a residue-numbering collision that must not be glossed.** *measured (mouse); this case UNKNOWN → S1/S2/S3* · Sieben 2020, PMID 31738183, Results/Figs. 8A, 9A, 6E; erratum PMID 33136097

4b — Raising BUBR1 is a validated intervention point in this disease, not an inference. In cells from biallelic MVA patients — low BUBR1, impaired checkpoint, chromosome-alignment defects — ectopic expression of BUBR1 *restored mitotic checkpoint activity*, proving BUBR1 dysfunction is what causes the segregation errors in patients. *measured (MVA patient cell lines)* · Suijkerbuijk 2010, PMID 20516114

5 — Downstream: a weak SAC → premature anaphase and lagging chromosomes → aneuploidy and micronuclei → mosaic variegated aneuploidy; p16/p19 senescence and progeroid pathology; predisposition to Wilms tumour, rhabdomyosarcoma and leukaemia. *measured (mouse + case series)* · Baker 2004, PMID 15208629; Baker 2008, PMID 18516091; Baker 2011, PMID 22048312; Hanks 2004; Ochiai 2019, PMID 31081598; Nishitani-Isa 2019, PMID 31184400

5b — A second, non-mitotic branch. BubR1 is also required for primary-cilium formation, via CDC20 degradation in G0 and APC/C-CDH1 control of Dishevelled. Morpholino knockdown in medaka causes cerebellar defects and perturbed left–right asymmetry, and the authors reframe PCS/MVA as *also* a ciliopathy — consistent with the Dandy-Walker complex and polycystic kidneys seen in patients. *measured (fish, cell)* · Miyamoto 2011, PMID 21389084

6 — Dose–phenotype map. A graded BubR1 series shows that ~10% of normal BubR1 (H/H) already produces progeroid disease; Sieben 2020 shows allele effects beyond abundance and aneuploidy, and even monoallelic carriers develop selective late pathology, with sarcopenia correlating with mTORC1 hyperactivity. *measured (mouse)* · Baker 2004; Sieben 2020

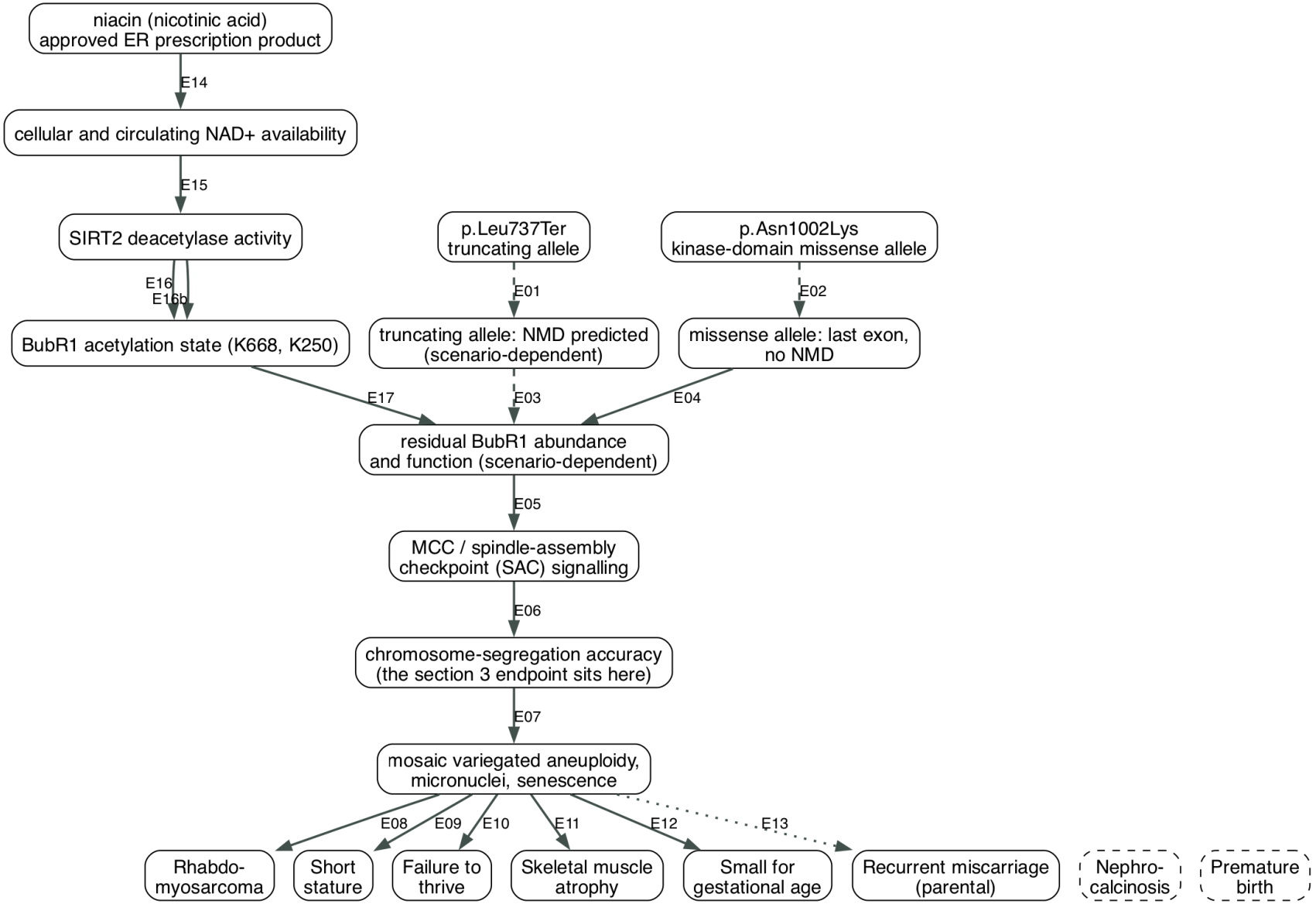
7 — A druggable lever. NAD+ → SIRT2 deacetylates BubR1 at K668 (opposed by the CBP acetyltransferase) → BubR1 is stabilized; NMN raises BubR1 abundance in vivo; a SIRT2 transgene extends the lifespan of BubR1 H/H mice (median +58%). Carried on the mechanism graph as edges E14–E17, each with its own strength, genotype match and counter-evidence. **Joint 3 below: this step has the furthest to travel of anything in the chain.** *measured (mouse, cell); substrate class differs from this case* · North 2014, PMID 24825348

7a — And that lifespan result is sex-dependent, which we had not recorded. North 2014's own abstract states the SIRT2 transgene increased median lifespan "with a greater effect in male mice". A third verification round on 2026-10-20 re-read the abstract and found this bound missing from every deliverable, including the sentence above and the pitch script. It matters more here than it would in a cohort study: the single load-bearing result of the whole therapeutic axis is sex-dependent, and an n-of-1 patient has one sex. We state it as a bound on the transferability of the lifespan endpoint, not as a prediction about this child, because the paper reports an effect size difference and not a mechanism for it — and the endpoint our decisive experiment measures is BubR1 abundance in cells, not lifespan, so the sex-dependence bounds the *motivating* result rather than the proposed readout. *measured (mouse); sex-dependent effect size* · North 2014, PMID 24825348

7b — That lever is specific to a BubR1-low background. The same group later tested SIRT2 overexpression on a normal, non-progeroid background and found no effect on health or lifespan, concluding that "strategies to enhance SIRT2 protein levels may not lead to increased longevity". This is not a refutation of link 7; it narrows it to exactly the setting this child is in, and removes any general-longevity halo from the argument. *measured (mouse)* · Wu 2023, PMID 38009412

7c — The same enzyme acts on a second BubR1 lysine — but the effect on abundance was not observed. Acetylation at K250, catalysed by PCAF, *blocks* ubiquitin-dependent degradation of BubR1, and SIRT2 deacetylates K250 in vitro and in vivo. The qualifier is decisive, in the source's own words: knocking SIRT2 down raised K250 acetylation, but that increase was "not substantial to elevate the levels of total BubR1 or delay the transition from prometaphase to anaphase". An established enzyme–site relationship, then — not an established opposing effect on abundance. **Joint 4 below.** *measured (human cells, both steps)* · Choi 2009, PMID 19407811; Suematsu 2014, PMID 25285631

8 — In humans, niacin escalated from 250 mg/day by 250 mg every four weeks to 750–1,000 mg/day (14–18 mg/kg; the 750 mg cap applied to the lightest subject) raised blood NAD+ up to 8-fold, and muscle NAD+ in patients reached the level of their controls — normalisation, not a measured fold-rise in muscle. *measured (adult)* · Pirinen 2020, PMID 32386566



The mechanism chain as the engine holds it. Eighteen edges, each carrying its own evidence strength, model system, allele-class match and counter-evidence. **The line style is the evidence grade**: a solid line is measured in a model system, a dashed line is computationally predicted, and a dotted line is asserted by a source rather than measured — so E13, the human half of the recurrent-pregnancy-loss edge, which its source states as a *prediction*, is visibly weaker than the rest, and the two predicted transcript calls are visibly not measurements. **A dashed box with no incoming arrow is a registered phenotype the chain does not explain**, left unconnected on purpose rather than joined by an arrow we cannot source. The therapeutic axis (top left, E14–E17) enters at the same node the disease axis bottlenecks on — residual BubR1 — which is the whole hypothesis in one picture, and why §8 turns on whether that node is an abundance problem or a function problem. **One thing the picture could be misread as saying:** E16 and E16b run in parallel into

the acetylation node, and they are not two supporting routes to the endpoint. E16 is the K668 step the hypothesis rests on; E16b is an established enzyme–site relationship at K250 whose own source found no change in total BubR1, so it is on the graph to show the breadth of the axis, not to add redundancy. Read the chain as a series circuit with one route, which is what §8b's leave-one-source-out finds.

The four joints where this argument can break

Every chain has a weakest link; this one has four, and naming them precisely is more useful than a confidence score. Each is an edge on the mechanism graph with its own counter-evidence field (`MECHANISM.md`); what follows is why each matters for the conclusion.

1. The class assignment for this child's allele is an inference, not a measurement. Suijkerbuijk 2010's abstract runs abundance-class ⇒ kinase-domain location. We need the converse, and the paper does assert it — but in the body, not the abstract: the Results heading reads "All kinase domain-localized mutations in BUBR1 reduce overall BUBR1 protein abundance" (Fig. 3, repeated in the Discussion). That universal is quantified over the **five** kinase-domain substitutions from British MVA patients the paper tested, not over arbitrary novel variants, and Fig. 2A shows an N-terminal TPR-domain substitution lowering abundance as well, just less steeply (2–6-fold across all patient lines) — a gradient, not two clean bins. Link 1c cuts the other way entirely: the same domain is independently required for PP2A-B56 recruitment, so a substitution inside it can produce a *function* lesion with no abundance change at all. The class assignment is therefore **unmeasured**, and §8 treats it as the pivot it is.

One human data point was added to this joint in 2026, and it is reported here because it runs in the direction this chain needs — which is the reason to state its bounds rather than its headline. Wei 2026 (PMID 42434306) identified two novel *BUB1B* missense variants in the probands of two unrelated families with unexplained recurrent pregnancy loss. Chromosomal microarray of six miscarriage specimens showed random aneuploidies; the probands' lymphocytes showed premature chromatid separation significantly above twenty controls by G-banding and centromere PNA-FISH; and Western blot showed a "trend toward reduced" BUBR1 protein, which the authors call preliminary functional support for haploinsufficiency. It is the first protein-level evidence for a *BUB1B* missense allele from outside Suijkerbuijk's British panel. **It does not close this joint, for a reason specific to this case:** p.W722R and p.A739S both sit N-terminal to the kinase domain (residues 766–1050) rather than inside it, so the paper tests whether a missense *near* the domain lowers abundance — not whether a missense *inside* it belongs to the abundance class, which is the question this case turns on. The remaining bounds: both probands are heterozygous rather than biallelic, both variants are VUS under ACMG/AMP, n = 2, and the protein result is a trend rather than a quantified fall.

2. The mouse allele that looks most like this one is a different residue. The allele in Sieben 2020 is BubR1L1002P, which the authors state models **human BUBR1 L1012P** ("human L1012 corresponds to mouse L1002"). This child's missense is at **human residue 1002** — a different residue. Nothing in that paper transfers to it directly. Three further bounds: the PCS finding is in the Results (Fig. 8A), not the abstract, whose summary sentence reads the other way (several progeroid pathologies were *attenuated* in the L1002P compound); the matched-abundance comparison holds for MEFs, while the splenocyte data (Fig. 9A) sit in the one tissue where BubR1 levels did differ significantly, against historical rather than contemporaneous comparators; and the direction is not simply "L1002P is worse" — median lifespan was 343 days for BubR1 H/L1002P against 196 days for BubR1 H/H.

3. The therapeutic lever has never been tested on this substrate class. The caveat is not merely "not tested with niacin". *No* NAD+ precursor has been tested on a BUB1B-mutant background. The in vivo NMN work was done on wild-type mice and WT / Sirt2-/- MEFs. The lifespan result is in BubR1 H/H, whose residual protein is **low in abundance but otherwise normal**, whereas this child's residual protein would be a misfolded missense product — a different substrate class. No E3 ligase is named in that work, so we cannot tell whether the degradation route K668 acetylation protects against is the same route that clears a misfolded kinase-domain mutant. And the K668 axis has no independent replication: a PubMed search on 2026-10-19 for `(BubR1 OR BUB1B) AND (K668 OR "lysine 668")` returns that paper and nothing else.

4. The enzyme we are proposing to recruit acts on BubR1 at more than one site — and we got this wrong twice before getting it right. SIRT2 deacetylates K668, which stabilises BubR1, and it also deacetylates K250, where acetylation blocks ubiquitin-dependent degradation. The obvious worry is that the second action opposes the first. **That worry is not supported by the evidence.** In the only study of that site, SIRT2 knockdown raised K250 acetylation but the increase was "not substantial to elevate the levels of total BubR1 or delay the transition from prometaphase to anaphase".

We record how this entry moved, because the movement is the point. An adversarial review raised K250 as a counter-argument. We could not obtain the source text at the time, so we graded it `asserted` and cited it for direction only — and, writing it up, described it as an opposing effect on protein stability, which over-states what the source says. A

later pass obtained the abstract. We had been wrong in both directions at once: under-grading the evidence, and over-stating the objection. **The correction runs in our own favour, which is precisely why it is written here rather than quietly applied.**

What survives is narrower and still real: SIRT2 has at least three mapped BubR1 lysines; their consequences for the aneuploidy endpoint do not line up (§9); and the net effect in a BubR1-low, misfolded-missense background has never been measured. For completeness on the published extent of this axis: it has exactly one review (PMID 39490401, *Semin Cancer Biol* 2024), which treats SIRT2 as an upstream regulator of BubR1 and names the interaction a therapeutic avenue for age-related disease and cancer. It is secondary literature and adds no primary measurement, so no claim here rests on it; it is cited so that a reader can see the whole of the axis we are working on rather than a selection from it. That is why the decisive experiment reads both sites as *fractions of total BubR1* rather than as raw signals (§6).

The **therapeutic objective** is a measurable endpoint, not "improve the disease": *restore accurate chromosome segregation by raising functional BubR1 toward the threshold where the SAC engages* — read out as BubR1 abundance/K668 acetylation, mitotic timing / nocodazole arrest index, and the aneuploidy/micronucleus rate (§6). Different candidate classes serve different endpoints and are not pooled.

Two endpoint caveats follow directly from the mechanism and are applied throughout: (i) **The aneuploidy rate is a readout, not the primary efficacy endpoint.** Across the allelic series, severity does not rank cleanly with the proportion of aneuploid cells (Sieben 2020 shows allelic effects beyond both protein level and aneuploidy), so a change in aneuploidy percentage alone neither proves nor refutes benefit. (ii) **Not every feature is expected to track aneuploidy at all.** Our own causal chain makes this concrete: of the eight registered phenotype terms, the segregation-error mechanism supports a sourced edge to six (the tumour, the growth and muscle features, one perinatal feature, and the parental reproductive observation) and **two are left explicitly unexplained** rather than connected by assertion — a renal feature and one perinatal feature. `MECHANISM.md` lists those two under *Phenotypes not explained by the current chain* (the generator's `unexplained_phenotypes` field), which is a result, not an omission. Note that one of the six is weaker than the rest: the human half of the parental-reproduction edge (E13) is the source authors' own *prediction*, not a measurement, so that edge is graded `asserted` — the lowest strength on our scale — rather than measured. If part of the phenotype arises from the ciliary branch (link 5b), those features will not improve with any aneuploidy biomarker, and an efficacy panel that measures only segregation would miss them. Because the proposed lever raises BubR1 *abundance* — which both the mitotic and the G0/ciliary roles depend on — the hypothesis is not confined to the mitotic branch; but any claim about the ciliary features requires its own readout, which this submission marks as pending rather than assumed.

4. Screening method: directionality, four patient gates, input validation

The screen is a small, auditable, disease-agnostic engine (`src/triage.py`, run by `run.sh`). It computes a gate verdict only; the scientific disposition is a separate human judgment in the canonical table, and the two are shown side by side (§5, `computed_vs_human.md`) so the tool's output never masquerades as science.

1. **Directionality prerequisite.** For a defect characterized as checkpoint insufficiency, a candidate must act to *strengthen* the checkpoint. Mitotic/checkpoint inhibitors — the disease's own direction — are auto-rejected.
2. **Four patient constraint gates**, each driven by a value from the case manifest: oncology (malignancy status), pediatric (age group), renal, growth. Three of them can reject; **the growth gate only raises a flag**, so we no longer describe all four as "hard gates" — one of them cannot stop anything. The gates read a candidate's `properties[]` tags and nothing else, which has a consequence we report rather than paper over: a candidate whose own `safety_assessment` records an axis as *contradictory* or *not assessed*, but which carries no property tag for that axis, **passes that gate without the gate ever seeing the risk**. Our internal cross-check only ran the other way, so this direction went through silently. The engine now computes those **gate-silent axes** for every candidate and prints them beside the verdict; `build_views.py` refuses to build unless the canonical entry declares exactly that set (`safety_gap_declared`); and `computed_vs_human.md` carries them as a column. For the named candidate they are oncology (contradictory) and growth (not assessed) — which is also why niacin can never reach the engine's top tier. No verdict changed when this was added.

3. **Objective classes, so a scenario can change the answer.** Each candidate declares which therapeutic objective it claims to serve (restore / mitigate / deselect / protect). The same drug can be the right direction for one objective and the wrong direction for another, so collapsing them into a single "strengthen/weaken" verdict hides the question. Declaring the class also makes the residual-function scenarios bite: §8 shows the named candidate surviving one scenario, flagged in another and rejected in the third. A candidate that declares no class blocks under a scenario rather than passing.
4. **Input validation that blocks rather than guesses.** Unknown fields, out-of-range values, non-boolean booleans, and a safety flag that contradicts its own gate (e.g. a "nephrotoxic" property with renal "not observed") all return BLOCKED, never a silently cleaned pass. Two input-laundering bugs found in internal review, and the two gate-reporting blind spots found in adversarial review on 2026-10-19, are now regression-tested, so the engine demonstrably refuses over-confident inputs — itself a rigor signal in an N-of-1 setting where data are sparse and the cost of false confidence is high. Twenty-eight self-tests pin this behaviour: five cover the scenario gate, six more cover malformed scenario blocks, and five run against **this patient's own constraint levels** rather than a permissive default — because until the 2026-10-19 review every self-test ran on an `adult / none / normal / normal` case, where three of the four gates carry empty rule lists and so could not reject anything. The `reject_if`, `require` and `flag_if` branches had no coverage at all; they do now.

A gate limitation we are reporting rather than patching. Reading the sirolimus label during eligibility verification turned up a boxed warning for immunosuppression with "possible lymphoma and other malignancies". Our G1 oncology rule rejects `immunosuppressive` only when malignancy status is *active*; at this patient's level — `history` — the reject list is `pro_proliferative`, `pro_tumor`, `increases_CIN`, which does not include it, so the gate lets sirolimus through. In a cancer-predisposition syndrome we judge that to be a real concern. **Sirolimus is not the only candidate affected**, and naming one example would make a structural property of the rule set look like a one-off. The engine derives it instead: any candidate carrying a risk tag that the active level's rule does not act on is printed as an `unactioned-risk-tag` note. On this case that is sirolimus (immunosuppressive, and growth_suppressive which only flags) **and** the dasatinib + quercetin senolytic pair (immunosuppressive) — a senolytic TKI combination passing the oncology gate of a child with a malignancy history is exactly the kind of thing a reader should be able to see without reading the YAML. We still did **not** rewrite the gate rule after seeing which candidates it would catch. Changing a scoring rule once you know its effect on a specific candidate is rule-fitting, and it would invalidate every other verdict the rule produced. What we changed is the engine's reporting, not its scoring: the verdicts are identical before and after. The rule limitation itself is logged for the next revision of the rule set.

A candidate reaches the engine's STRONG tier only with eligibility ELIGIBLE, evidence distance ≥ 3 (a rescue in disease-relevant cells), rescue = true, and every safety gate explicitly assessed with no unresolved risk. By design, no current candidate reaches STRONG — that is the honest result (§10), not a tuning failure, and self-test 12 proves STRONG is still reachable by a clean input.

5. Candidate funnel — one denominator, eligibility separated from scientific judgment

All candidates live in one canonical source (`candidates_canonical.json`); every table and the engine fixture are generated from it, so a synonym or a negation in prose can never change a structured verdict. Denominator: **13 interventions + 3 mechanism-branches + 1 negative-control group = 17 entries**, computed once from the canonical source and checked against every deliverable by `make_manifest.py`. The patient layer is `fact_register.csv` (18 registered facts: 7 CONFIRMED, 2 ASSUMED, 9 UNKNOWN); the candidate layer is in `candidate_table.md` (auto-generated) with this shape:

- **Eligibility** (a regulatory fact about a product) and **research judgment** (a scientific disposition) are kept in separate columns. A molecule can be ELIGIBLE and still INSUFFICIENT.
- The strongest-biology branch, raising BubR1 via SIRT2 (**C-NAD**, NMN/NR), is **INELIGIBLE** as a Track-2 medicine — NMN and NR are supplements, not approved drugs, and nicotinamide is both a supplement and directionally wrong (a sirtuin inhibitor that *lowered* BubR1 in North 2014). Its approved-product instance is evaluated separately as **C-NA** (niacin), the named candidate.

- Our earlier lead, **4-PBA** (C-PBA, sodium/glycerol phenylbutyrate), is held at **INSUFFICIENT** and is demoted from the named candidate consistently across report, pitch, candidate table and submission record: its mechanism is secretory-cargo export, not a cytosolic chaperone, with zero BUBR1-specific data, and its achievable exposure overlaps its HDAC-inhibitory range so an effect could not be attributed.
- Mitigation/deselection candidates (NAC, folate+B12, rapamycin, AMPK-deselection) are retained as CONDITIONAL for named endpoints other than raising BubR1; senolytic and mitotic-inhibitor entries are the wrong-direction negative controls. The mitigation lane was the weakest-supported one in this submission, resting on general oxidative-load reasoning rather than on anything measured in an aneuploidy model. That changed in 2026, and we record it as support for *the lane* and not for a molecule: in a *Drosophila* MVA model made by depleting a single SAC gene in the neural-stem-cell compartment, proteostasis failure and mitochondrial dysfunction were shown to mediate how complex aneuploidy damages stemness, and overexpressing ROS scavengers or mitochondrial chaperones, or blocking apoptosis, dampened the aneuploidy-driven loss of brain size (PMID 41820377, *Nat Commun*). Bounds: *Drosophila*, genetic manipulation rather than a drug, an organ-size endpoint, and NAC itself untested there. **A second result does test the molecule, and we had missed it** (added 2026-10-20 by a search run against our own global negatives): in Sod1-deficient mouse oocytes matured *in vitro*, chromosome misalignment and segregation errors rose alongside **reduced recruitment of BUBR1 to kinetochores**, and adding N-acetyl-L-cysteine or L-ascorbic acid during maturation reduced the frequency of segregation errors (Nago 2025, *Reprod Med Biol*, PMID 39845481). That is a drug-level, mammalian result on a segregation-fidelity endpoint, where our write-up had said the lane rested on general oxidative-load reasoning. Its bounds are what keep the verdict where it is: an oxidative-stress knockout rather than MVA or *BUB1B*, a BUBR1 lesion of kinetochore recruitment rather than low abundance, meiosis in oocytes rather than mitosis in somatic cells, and a drug added to maturation medium rather than a systemic exposure in a child. So the lane is better supported than we said, and the question that makes N-acetylcysteine CONDITIONAL rather than supported is unchanged: whether it reaches a useful exposure in a child.

Why niacin rather than the nearest alternatives

The engine marks several candidates CONDITIONAL. That is a gate verdict, and a gate cannot rank — it only filters. The ranking below is a **human judgment**, and we set it out explicitly so it can be argued with, rather than letting "the engine chose it" do work it cannot do. Only `restore`-class candidates are comparable here; mitigate and deselect candidates serve different endpoints and are not competing for the same experiment.

	Niacin (C-NA)	**NMN / NR (C-NAD)**	**4-PBA (C-PBA)**
Mechanistic distance to BubR1	Two steps, both with primary sources: NAD+ → SIRT2 activity → K668 deacetylation → abundance	**Identical** — same axis, same evidence	Far. Secretory-cargo export via COPII sorting; **zero BUBR1-specific data**
Achievable human exposure	Measured in adults: 750–1,000 mg/day raised blood NAD+ up to 8-fold	Not measured in a comparable dosing study	Measured, but for urea-cycle indications unrelated to this axis
Functional endpoint it would move	BubR1 abundance, then checkpoint function	Same	None specified that connects to BubR1
Pediatric constraint	No pediatric label; off-label pediatric series exists, with 38% discontinuation	Not an approved drug, so the question does not arise	**Best of the three** — pediatric label from under 2 months
Oncology risk	Unresolved (contradictory); NAD+ supply is pro-tumour in some models	Same unresolved axis	No specific signal
Why it is or is not the pick	**Chosen.** The only molecule on this axis with an approved product and a dated label that a clinician could actually prescribe	**Not eligible.** Supplements, not approved drugs — this is a Track-2 eligibility fact, not a scientific one	**Not chosen.** Demoted on *mechanism*, not eligibility. Its achievable exposure also overlaps its HDAC-inhibitory range, so an effect could not be attributed

What would change this ranking. 4-PBA overtakes niacin if any BUBR1-specific chaperone data appear, or if an MVA cell line shows BubR1 rising under 4-PBA at exposures below its HDAC range. NMN or NR overtake it if either gains an approved product in a relevant jurisdiction. Niacin is retired outright by Stage B of §6 if the K668-acetylated

fraction does not fall, or by Stage A if intracellular NAD+ does not move.

The honest bound on this comparison. It ranks three candidates we reached by literature-driven search. It does not establish that niacin is the best approved medicine for this child, because we did not run a systematic screen that could have surfaced a non-obvious fourth (§13).

6. Surviving candidate and the decisive experiment (which is also diagnostic)

The named candidate **niacin** survives the rule as **conditional**. One experiment decides it, and the same measurement has standalone value: it would functionally characterise this child's residual BubR1, readable against the Sieben 2020 dose × allele-quality series. **Bounded deliberately:** that series is a mouse allelic series, so this is a research-grade characterisation, not a validated diagnostic or prognostic instrument.

Material (held by the organizers / clinical team; this team does not access it): patient-derived fibroblasts or a lymphoblastoid line; an isogenic (CRISPR-corrected) or healthy control; a BubR1-knockdown positive control (siRNA titration to an H/H-like level).

The experiment is not starting from zero: **ectopic BUBR1 has already been shown to restore checkpoint activity in cells from biallelic MVA patients** (Suijkerbuijk 2010), so a positive control for "more BUBR1 fixes the checkpoint in this disease" exists in the literature. What is untested is whether a *drug* can raise endogenous BUBR1 on this allele background. That single step is what the dose-response arm below measures, and it is the whole distance between the published rescue and a repurposing hypothesis.

The experiment is staged, and the stages exist to tell failures apart. A single-stage design has a fatal reading problem: if BubR1 does not rise, that is at least five different results — the drug never reached the pathway, intracellular NAD+ never moved, SIRT2 was not the limiting step, this mutant protein cannot be stabilised at all, or it was stabilised and function still did not recover. Each one sends follow-up somewhere different, and only the fourth and fifth say anything about the hypothesis. So each stage below answers one question, and a negative at any stage names which of those five it is.

Stage A — did the drug reach the pathway at all? Intracellular NAD+ and the relevant metabolites (NAD+, NMN, nicotinamide, and the 4PY/2PY terminal metabolites) by LC-MS, in patient and control cells, across the dose–response. *If intracellular NAD+ does not move, nothing downstream is interpretable.* That is an **exposure and metabolism** result, not a mechanism result: the next step is a different precursor or a different exposure, not abandoning the axis. Reporting a null here as "the hypothesis failed" would be the single easiest way for this submission to mislead, so Stage A runs first and gates the rest.

Stage B — did the pathway act on the target, and through SIRT2? Total BubR1, and the **acetylated fraction at K668 and K250 normalised to total BubR1**, by IP with site-specific antibodies. **Direction, stated explicitly because it is easy to get backwards:** SIRT2 *de*acetylates K668, so a working intervention makes the K668-acetylated **fraction fall** while total BubR1 rises. Normalising matters — an unnormalised site signal moves with abundance and can be read either way. Run with and without **SIRT2 knockdown (siRNA) and a SIRT2 inhibitor**, because that arm is what makes the result attributable:

- NAD+ rose, K668 fraction fell, BubR1 rose, **and the rise is abolished by SIRT2 knockdown** → the proposed route is operating in these cells. This is the only result that supports the stated mechanism.
- NAD+ rose, BubR1 rose, **but SIRT2 knockdown does not abolish it** → BubR1 went up by some other NAD+-dependent route. Potentially useful, but it is not this hypothesis, and the report would have to say so.
- NAD+ rose, K668 fraction did not fall → **SIRT2 is not the limiting step here.** The axis may be real biology and still not actionable in this background.
- K668 fraction fell but total BubR1 did not rise → **this mutant protein is not stabilisable by deacetylation.** Scenario S1 is retired; abundance-raising strategies go with it.

The one reagent in this design that is not off the shelf, named rather than assumed. Stage B is the stage that makes the result attributable, and it is written above as "IP with site-specific antibodies" as though an anti-acetyl-K668 BubR1 reagent were a catalogue item. A supplier search on 2026-10-20 (Cell Signaling, Abcam, Thermo, Boster, LSBio, Affinity and the Biocompare aggregator, query terms "acetyl-K668 BUB1B / BubR1 acetyl lysine 668") returned **total**-BubR1 antibodies and no site-specific acetyl-K668 product; the only acetyl-BubR1 monoclonal we located is a patent filing whose target lysine the record does not state. *This is a bounded supplier search, not a global*

negative: we did not contact suppliers, and a custom-catalogue item may exist. North 2014 used reagents raised for that study, so the realistic routes are a material transfer from the originating laboratory, a custom acetyl-peptide antibody, or — preferably, because it needs no new reagent and quantifies both sites at once — an **antibody-free readout: targeted mass spectrometry (PRM) on immunoprecipitated BubR1, reporting acetyl-site occupancy at K668 and K250 as a fraction of the corresponding unmodified peptide.** That substitution does not change what Stage B asks or how it reads. Every other measurement in Stages A–E (LC-MS metabolites, total BubR1 by Western, nocodazole arrest and washout by live imaging, missegregation per division, barcoded clonal fate, N-terminal truncation blot) uses standard assays and commercially available reagents, so this is the design's single procurement risk and it has a stated fallback. A reviewer who wants one number for feasibility should read this paragraph instead: we do not state a cost or a timeline, because we would be inventing both.

Stage C — does more protein restore function? Checkpoint **engagement and silencing**, not arrest length alone: nocodazole arrest index *and* time to mitotic exit after washout, by live imaging. A longer arrest is ambiguous on its own — it can mean a recovered checkpoint, an inability to silence the checkpoint, or toxicity, and those are opposite news. Alongside it, **new missegregation events per division** by live imaging, which is the quantity the mechanism actually predicts, rather than the steady-state aneuploid fraction, which is a history. *If abundance rises on-route and function does not recover, that is scenario S2: a per-molecule function lesion. Abundance-raising is retired for this child; mitigation and deselection remain.*

Stage D — is any improvement repair, or is it selection? Absolute cell counts, death and proliferation, plus clonal fate (barcoding or continuous tracking), in the same wells and at the same timepoints. The criterion is the aneuploid **fraction** of the surviving population, never raw survival: a genuine checkpoint rescue lowers the rate at which new aneuploid cells arise and may well *raise* overall survival. A fall in aneuploid fraction accompanied by a fall in absolute normal-cell number is **selection, not repair**, and must be reported as such.

Stage E — the S3 discriminator. Truncated product by N-terminal antibody. If a truncated product is present and rises with NAD+ repletion, every abundance-raising strategy pauses until that is characterised.

What this experiment cannot settle, stated plainly. A clonal-fitness arm in cultured cells can surface a risk signal; it cannot establish long-term tumour safety in a cancer-predisposition syndrome. Nothing here is a safety clearance, and §9's oncology item stays open regardless of the result.

Rules: ≥ 3 independent biological replicates, blinded scoring, and death/proliferation measured alongside every "improvement" so a reduction in aneuploid cells by selection is never read as repair. The report states what resources are required and whether they are obtainable; it does not promise a timeline or cost. No dose is specified; the in-cell concentration range is written as *pending in-cell dose–response*.

7. Eligibility matrix — product level, with dated sources

Approval is a property of a *product*, not a molecule, so each row is a product (molecule × form/route × jurisdiction × indication × age range × status × checked-date × an openable source URL × any counterexample). Generated from `docs/eligibility_matrix.csv` (public regulatory data, committed) into `docs/eligibility_matrix.md`. Two rows carry weight beyond their own candidate. First, **the two niacin products are not interchangeable**: the immediate-release label states that sustained-release preparations "should not be substituted for equivalent doses of immediate-release (crystalline) nicotinic acid", and that severe hepatic toxicity including fulminant hepatic necrosis has occurred on such substitution. Any niacin hypothesis must name the formulation; "niacin" alone is not a specification. Second, **4-PBA's demotion is not an eligibility problem**: glycerol phenylbutyrate has the broadest pediatric label of anything in the table (established from under 2 months), and it was demoted purely on mechanism. Conversely, ataluren shows why eligibility has to be read per jurisdiction, and we had this wrong. The EU conditional authorisation was **not renewed on 28 March 2025** because effectiveness was not confirmed, and the US application was withdrawn on 2026-02-12, from which we concluded that no approved readthrough agent exists. It does: ataluren retains a **Great Britain** marketing authorisation, with three live Translarna SmPCs on the UK electronic medicines compendium, the 250 mg one updated 31 March 2026 (checked 2026-10-20). One search breaks a global negative, and this one broke ours. What survives is narrower and is an eligibility statement rather than a claim about the world: the only approved readthrough product is approved for a different gene and a different indication, so reaching this child's truncating allele with it would need an off-label argument that this submission has not made, and the approved aminoglycosides

are rejected on renal grounds. Whether that off-label route should put the readthrough branch back in the eligible set is a scientific judgment we have **not** made here; the branch stays INELIGIBLE and the open question is named instead of being settled quietly.

The load-bearing row: niacin extended-release tablets are an approved US prescription product (DailyMed label revised 12/2025, checked 2026-10-19), indicated for dyslipidemia in adults; the label states that "*Safety and effectiveness of niacin therapy in pediatric patients (≤ 16 years) have not been established.*" That is precisely why the research judgment is CONDITIONAL and the pediatric path is off-label and specialist-supervised (Colletti 1993), not an approved indication — and note that the pediatric experience we are relying on for feasibility is the same series that reports a 38% discontinuation rate, at doses that overlap what this hypothesis would need (§9).

8. Scenario sensitivity — three residual-function hypotheses, three engine runs

Because residual BubR1 function in this child is UNKNOWN (§2), the engine is run on three case files (generated by `scripts/phase1/14_build_scenarios.py`), and the three verdict sets are shown side by side — turning "we don't know" into an analytic output rather than a guess. Each scenario declares which therapeutic-objective classes remain valid, and the engine enforces that: a candidate declares the objective class it claims to serve, and a class marked `invalid` in the active scenario is rejected, `conditional` is flagged. Running the same candidate table against the three scenario case files gives three different answers:

Scenario	Surviving (CONDITIONAL)	Rejected	The named candidate
S1 stabilizable hypomorph	7	10	**survives** — niacin CONDITIONAL
S2 misfolded, non-functional	5	12	**overturned** — rejected at `objective-validity[S2]`, together with every other restore-class candidate
S3 truncation escapes NMD	7	10	**survives but flagged** `objective-restore-conditional[S3]`

The base case carries no scenario block, so the gate is inert there and the base result is unchanged — the sensitivity is an added analysis, not a silent re-scoring.

Scenario	Hypothesis	Consequence for the named candidate
S1 stabilizable hypomorph	missense protein present but unstable; truncating allele decayed	NAD+/SIRT2 stabilization is on-direction; **niacin is CONDITIONAL**
S2 function lesion	missense protein, even if stabilized, lacks checkpoint function — **directly supported for this domain** by Gama Braga 2020: the pseudokinase domain is required for PP2A-B56 recruitment, and mutating it causes alignment errors independent of abundance	raising abundance cannot help → **niacin overturned**; only mitigation/deselection remains; the reportable gap (§10) becomes the main result
S3 truncation escapes NMD	a truncated product exists, possible dominant-negative	raising total BubR1 may also raise the truncated product → any strengthen-direction strategy pauses until the truncated product is measured

The named candidate holds **only under S1**; S2/S3 are discriminated by the first two readouts of the decisive experiment. This dependency is the point, not a weakness. **What it is not is a probability, and we withdraw an earlier claim that it was.** `config/scenarios/bub1b_mva.scenarios.yaml` assigns the three scenarios equal weight (0.333 / 0.333 / 0.334). An earlier draft of this section read that as "about a one-in-three chance of being directionally invalid". That inference does not hold and we retract it: the three scenarios are **not mutually exclusive and not exhaustive**. S1 and S2 describe the *missense* allele's protein — present but unstable, versus present but non-functional — while S3 describes the *truncating* allele escaping NMD, which can co-occur with either. Real configurations we have not enumerated (partial NMD escape with partial function loss, for instance) sit outside all three. Equal weights are therefore a **stress-test convention** — each branch is exercised so none is quietly assumed away — and nothing more. A defensible prior would need the axes separated (NMD outcome, protein abundance,

per-molecule function, metabolic entry, risk) and evidence on each; we have none, so we assert none. What the analysis does show, and this is the part that stands, is that the named candidate depends entirely on one branch, and that the decisive experiment's first two readouts separate the branches directly.

One caveat on what this sensitivity analysis does and does not demonstrate. The three scenarios differ on a single axis — `objective_validity` for the `restore` class (valid / invalid / conditional); `mitigate`, `deselect` and `protect` are valid in all three. So the analysis tests exactly one thing: how much of the candidate set depends on the restore-class premise. The answer, two candidates of seventeen, is a real result about the funnel's concentration, but it is not a broad robustness check, and the `objective_class` assignment for each candidate was made in a single human pass and is recorded as a judgment, not a measurement. The broad check is §8b, which perturbs the engine along five axes rather than one.

8b. Did we build the engine to produce this answer?

§13 concedes that the decision rule in §1 was written alongside the candidate work rather than pre-registered before it. That concession invites the question a reviewer should ask of any repurposing engine: **were the rules shaped until they emitted the candidate already chosen?** It cannot be settled by reading the rules, because a fitted rule and an honest rule look identical on the page. It is settled by perturbing the engine and watching what moves. `scripts/phase1/20_robustness.py` does that on five axes and runs inside `run.sh`, so a result we would rather not publish cannot quietly fail to be published; the full tables are in `docs/ROBUSTNESS.md`. Two of the five are assertions that fail the build. Baseline for all of them: seven of seventeen candidates eligible, the named candidate CONDITIONAL.

Start with the result that does not flatter us. Leave-one-source-out on the therapeutic chain shows the chain is not a web of mutually reinforcing evidence but a **series circuit with two single points of failure**. Removing North 2014 (PMID 24825348, *SIRT2 induces the checkpoint kinase BubR1 to increase lifespan*) leaves edges E15 and E16 with no source at all; removing Pirinen 2020 (PMID 32386566, the niacin→NAD+ link in humans) leaves E14 unsourced. Either removal cuts every path from the intervention to the endpoint. One paper carries the entire SIRT2→BubR1 core. We report this instead of counting the chain's full citation list, which is longer and would imply a redundancy that is not there. **And the scope of that experiment is narrower than its headline, which we state rather than let a reviewer find.** It drops papers from the therapeutic chain only. Counting single-source edges across the whole mechanism graph gives five, not two: the two above, plus E05 — the patient-cell rescue, sole source Suijkerbuijk 2010 — and E09/E10 (the growth phenotypes, sole source Baker 2004) and E12 (sole source Schmid 2014). E05 is the uncomfortable one, because it is the edge that makes raising BUBR1 a *validated* intervention point in this disease rather than an inference (§3 link 4b), and a reader who doubts that one paper loses the disease-relevance premise as surely as a reader who doubts North 2014 loses the drug. Extending the harness to drop sources across the whole graph rather than one chain is the obvious next revision and has not been done. The practical consequence: a reader who doubts either paper should treat this hypothesis as **unsupported, not merely weakened** — and it is exactly why stages A and B of the decisive experiment (§6) measure those two links in the patient's own cells rather than inherit them from the literature.

Rejections are over-determined, which is the hard thing to fake. Testing each rejected candidate against every rule independently rather than stopping at the first one that fires: **nine of ten rejections are reached by two or more unrelated rules on their own**. Five are reached by three. Rule-fitting does not produce that pattern — tuning the oncology gate does not also arrange for the eligibility rule and the directionality prerequisite to agree about the same compound. The single exception is named rather than buried: **C-AMPK (metformin) is rejected by one rule only**, `G3_renal`, and the constraint perturbation confirms it — change this child's renal status to either `impaired` or `normal` and metformin returns to the eligible set. If the renal gate is wrong about metformin, the funnel is wrong about metformin, with nothing else to catch it.

Ablation also shows where rules are redundant rather than idle. Removing the directionality prerequisite changes no outcome, but not because it does nothing: the three candidates it rejects (`N-ASA`, `N-ISO`, `N-MITOTIC`) are then caught by the pediatric or oncology gate instead. The same holds for competition eligibility — remove it and all six ineligible entries are still rejected downstream. `G2_pediatric` is the broadest rule in the set, appearing in nine of the ten rejection lists, yet removing it frees nobody. An earlier version of this analysis diffed on the verdict alone and so reported "directionality changes nothing", which was false in the way that matters: redundancy is not inactivity, and the distinction is now in the output.

The engine does not know which candidate is ours. Every identifier and drug name was replaced with an opaque token and the list shuffled before the engine saw it: seventeen candidates, **zero verdict differences**. Six synthetic probes were then injected alongside the real ones, each with a known-correct answer. A twin of the named candidate — identical in every field the engine reads, under a different name — receives the identical verdict, which is the direct test for favouritism. A mirror of it, acting in the opposite direction, is rejected at directionality. Poisoned twins carrying `increases_CIN`, `nephrotoxic`, or no pediatric path are rejected at G1, G3 and G2 respectively; an anti-anabolic twin is flagged by G4 and not rejected, because G4 can only flag. **Six of six behaved as specified**, and the probes did not disturb any real candidate's verdict.

One observation ties these together. No gate ablation and no constraint perturbation moves the named candidate off CONDITIONAL — loosening a safety rule cannot promote it, because STRONG additionally requires rescue evidence in disease-relevant cells at `evidence_distance >= 3` and this candidate has neither. But switching the scenario to S2, in which the missense allele is a function lesion rather than an abundance problem, **rejects it outright**. The only perturbation that moves our own answer is the one that attacks the biology. That is the separation we want between a safety gate and an evidence claim, and it is checkable by anyone who runs the script.

What this does not establish. Perturbing an engine tests the engine, not the biology. R1–R4 would look exactly like this if every property value fed into them were wrong: the gates would still bite, the twin would still match, the names would still not matter. What they establish is narrower and worth stating exactly — the verdicts follow from the recorded fields by a rule that treats an unnamed compound the same as the named one. Whether those fields are true is the work of §3, the citation check and the claim ledger. And the probes test the gates along the axes we thought to mutate; a gate failure on an axis we did not think of would not appear here.

9. Counter-evidence (bounded, not withdrawn)

The strongest arguments against the primary hypothesis, each bounded to its own model and readout rather than extrapolated to a clinical contraindication:

- **The premise that NAD+ repletion will raise BubR1 here is not established, and this is the strongest objection to the whole hypothesis.** North 2014's logic is that BubR1 declines *in normal ageing* because NAD+ declines. In MVA, BubR1 is low for a structural reason — an absent transcript plus a destabilised missense protein — and **no study reports NAD+ status in MVA or in BUB1B-mutant cells** (PubMed, searched 2026-10-19, no hits). SIRT2 can only deacetylate protein that exists; a protein being turned over because it is misfolded may not be rescued by deacetylation at all. The decisive experiment's first readout exists precisely to test this step, and if it fails the hypothesis is dead regardless of everything else in this report.
- **The second strongest objection is that this allele may be a function lesion, not an abundance problem.** The named candidate works only if the missense allele yields protein that is *scarce but competent*. Gama Braga 2020 shows the pseudokinase domain is required for KARD phosphorylation and PP2A-B56 recruitment, and that mutating it produces checkpoint-silencing and alignment defects directly. A substitution inside that domain could therefore be functionally dead at normal abundance — scenario S2, in which the engine rejects niacin along with every other restore-class candidate. We do not claim the literature settles which class this allele is in. It does not, and we have no measurement of our own.
- **SIRT2 does not act on BubR1 at one site, and the sites do not agree.** Joint 4 (§3) sets out what the K250 evidence does and does not support — notably that the one study of that site found no change in total BubR1. A third site makes it worse rather than better. In mouse oocytes the *non*-acetylatable BubR1-K243R is the **beneficial** allele: it partly prevents the meiotic deficits of Sirt2 depletion and, in oocytes from aged mice, "lowers the incidence of aneuploid eggs" (Qiu 2018, PMID 29067790). Mouse K243 is plausibly the same conserved residue as human K250 under a different numbering, which we have not resolved. So: one enzyme, at least three mapped sites, consequences for stability and for the aneuploidy endpoint that do not line up, and no measurement of the net effect in a BubR1-low misfolded-missense background. Two bounds on how far we take this: Qiu 2018 is **meiosis in oocytes, not mitosis**, and its rescue comes from over-expressing a mutant BubR1, not from raising endogenous protein or from any drug. And one point that cuts for us rather than against, stated because leaving it out would be selective: the *direction* of that paper runs with this hypothesis, not across it — Sirt2 depletion damages the spindle, the acetylation-mimetic allele reproduces the damage, and the

deacetylated-state allele lowers the aneuploid fraction. The problem it creates for us is not direction but bookkeeping: one enzyme, at least three mapped sites, and stability and aneuploidy outcomes that do not line up site by site.

- **The largest randomised safety dataset for this exact product is a null trial with real harms.** HPS2-THRIVE (Landray 2014, PMID 25014686, N Engl J Med) randomised **25,673** adults with vascular disease to 2 g/day extended-release niacin plus 40 mg laropiprant or placebo. Over a median 3.9 years it "had no significant effect on the incidence of major vascular events (13.2% and 13.7% ... rate ratio, 0.96; 95% CI, 0.90 to 1.03; P=0.29)", with significant absolute excesses of serious adverse events: disturbances in diabetes control +3.7 percentage points, new diabetes diagnoses +1.3, infection +1.4, gastrointestinal +1.0, musculoskeletal +0.7, bleeding +0.7, skin +0.3 (all P≤0.003). **Bounds, stated so this is not read as more than it is:** the comparator contains laropiprant, so attribution to niacin alone is confounded; the population is adults with established atherosclerosis, not children; the endpoint is cardiovascular, not anything in this hypothesis's causal chain; and the drug was given for lipid modification, not NAD+ repletion. One feature cuts the other way and we note it rather than omit it — the trial ran a pre-randomisation run-in that **excluded people who could not tolerate ER niacin**, so the population is enriched for tolerators and real-world harm would plausibly be higher. This does not touch the mechanism argument. It does mean that if this hypothesis ever moved toward a human exposure, the tolerability prior is poor and well quantified.
- **NAD+ may fuel tumours**, which matters given MVA's cancer predisposition — but that evidence comes from tumour models, not a BubR1-deficient background, and raising BubR1 is itself anti-aneuploidy and tumour-suppressive (Baker 2013, 23242215). Marked *contradictory*; adjudicated by the decisive experiment's clonal-fitness arm.
- **The niacin metabolite 4PY** is associated with cardiovascular events (Ferrell 2024, 38374343) — in adult cohorts, not extrapolated to a pediatric setting; recorded, not projected.
- **Nicotinamide lowers BubR1** (North 2014, Fig 1) — bounded to the sirtuin-inhibiting concentration; the conclusion is "NAD+-pathway molecules are not interchangeable," not "the axis fails."
- **Pediatric niacin experience is not clean, and the dose context does *not* excuse it** (Colletti 1993, PMID 8516088): 21 children aged 4–14, "treated with niacin, 500 to 2250 mg daily" — adverse effects in 16/21 (76%), reversible dose-related transaminase elevation in 6 (29%), discontinuation in 8 (38%) for flushing, abdominal pain, vomiting, headache or transaminase rise. We previously bounded this to "lipid-lowering doses", which invited the reading that the exposure is not transferable. **It is.** That range brackets, and at its top exceeds, the 750–1,000 mg/day to which Pirinen 2020 titrated adults to raise NAD+ — so this is pediatric safety data *at the doses the hypothesis would need*, and it cannot be discounted on indication. Two real bounds remain: the series is retrospective, not a trial, and its efficacy figures are quoted only for the >1,000 mg/day subgroup. This is the basis for CONDITIONAL, not SUPPORTED.
- **Anemia tendency** (Pirinen 2020) — adults, 10 months; a monitoring item, not a contraindication.

None of these is softened, and none retracts the hypothesis; each is a line on the ledger with a disposition.

10. A reportable gap

The strongest link in the mechanism chain — directly stabilizing BubR1, or selectively activating SIRT2 — **has no approved drug**. The gap can now be stated precisely: the disease-relevant rescue (more BUBR1 restores the checkpoint in patient cells) is *published*; the pharmacology to achieve it endogenously is *absent*. Niacin is an indirect, systemic proxy for it. This is a finding, not an omission: it tells follow-up researchers to look toward SIRT2 activators and BubR1-K668 deacetylation, and away from secretory-pathway chaperones (the 4-PBA direction we demoted).

Two independent checks now stand behind that gap, and they are stronger than our original search. First, the obvious approved-drug routes to raising BUBR1 have been *tried and reported to fail*, not merely left unsearched: in acute myeloid leukaemia, where BubR1 is frequently repressed, **decitabine produced no significant BubR1 upregulation and the HDAC inhibitor trichostatin A did not stabilise it** — re-expression was achieved only genetically (Schnerch 2013, PMID 23812934). That covers the two routes a reader would reach for first, transcriptional de-repression and acetylation-mediated stabilisation. Second, the gap is visible at target level and not only in label text: Open Targets (BUB1B, ENSG00000156970, data 26.09, queried 2026-10-19) returns Approved Drug, Advanced Clinical and Phase 1 all false across small-molecule, antibody, PROTAC and other-clinical modalities, with no high-quality pocket and no ligand-bound structure; the one positive tractability signal is ubiquitination evidence supporting

a **degrader**, which is the opposite of the direction this report needs. A label-text search for "SIRT2 activator", which is what GN-03 originally ran, could never have found a drug raising BUBR1 by another route; these two checks close that recall hole. For this child and for others with checkpoint-insufficiency MVA, naming where the approved-drug space is empty is itself a contribution to how the disease is approached.

11. Scalability and boundaries

- **Already demonstrated:** the engine is case-driven — a synthetic gain-of-function case file flips the directionality rule with no code change (`config/cases/synthetic_gof_demo.yaml`). The same rule applies across MVA genes in `MVA_DECISION_TABLE.md`: for each gene, the defect direction, the first experiment, and whether the NAD+/SIRT2 axis is even relevant (it is specific to BUB1B).
- **Honest boundary:** the engine demonstrates gate logic; candidate generation and evidence grading remain human-curated. This is **not** a cross-disease discovery system, and we do not claim it is.
- **To move to another disease:** add a case file; if the defect direction is new, add one rule to `config/scoring.yaml`. That is the whole extension surface.

11b. Every load-bearing claim is on a ledger

Each statement in this report that affects the conclusion has a row in `registers/claim_ledger.csv` — 37 claims: 18 mechanism (one per link of the causal chain, carrying that link's own evidence strength and its counter-evidence — five of them, E14–E17 and E16b, are the **therapeutic** axis, which until 2026-10-19 was the one part of the argument carried by no ledger at all), 12 regulatory (one per product row of the eligibility matrix, each with a dated label URL), 5 global-negative, and 2 method. The ledger is *generated*, not hand-maintained: `make_claim_ledger.py` builds it from the mechanism chain and the eligibility matrix on every clean run, so a claim cannot drift away from its source.

Two deliberate properties. First, the five **global-negative** claims — the "no approved drug does X" statements that one counterexample would destroy — each carry a dated, reproducible search in `registers/search_strategies.md`, with the verbatim query, the hits and why each hit does not count. Two of the five are marked `unresolved` rather than `bounded`, because they concern the premise the named candidate most depends on. Second, the generator **refuses to emit any row matching a genomic-coordinate or genotype pattern**, and that refusal is exercised as a test: the working evidence ledger from early October contains coordinates, so it is excluded from this artefact by construction rather than by care.

12. Reproducibility

`run.sh` runs the self-test, generates all views from the canonical source, runs the real and synthetic cases and the three scenarios, and ends with `make_manifest.py`, all from pinned dependencies in a clean environment; it checks for PyYAML first and tells you how to fix a missing dependency rather than dying three steps in. `make_manifest.py` is the consistency gate: it recomputes the claim-ledger size, the global-negative count, the registered-fact count, the product-row count and the self-test count (by executing the self-tests, not by counting lines) from their sources, refuses to pass if any deliverable disagrees, and cross-checks that this report and `METHODS_AND_AI.md` disclose the same number of safety-classifier interruptions. `run.sh` also runs the robustness harness (§8b), whose name-blinding and probe assertions exit non-zero, so an engine change that made a verdict depend on a candidate's name would fail the build rather than the review. Because `ROBUSTNESS.md` describes the engine's behaviour, a stale copy of it would be a confident false statement about code that no longer exists; the manifest therefore blocks if it is older than `triage.py`, `scoring.yaml`, the candidate file, the case file or the mechanism graph, and blocks if it records a failed assertion. Unlike the citation check, this needs no network, so staleness here has no excuse and does not merely pend. **The gate is itself perturbed, not vouched for.** Until 2026-10-20 nothing tested `make_manifest.py`: the claim that the deliverables cannot drift rested on the gate having caught us in the past, which is an anecdote about a checker rather than evidence about it. `tests/phase1/test_manifest_gate.py` now copies the whole tree to a scratch directory, breaks exactly one thing, and requires the gate to block with the right message — a candidate count retyped wrongly in any deliverable, a single word changed in the PRIMARY sentence, a committed run snapshot that records a failed step, a pitch script pushed over the time limit, the pitch's own stated length

disagreeing with its computed length, an AI provider present in the usage register but absent from the disclosure, a provider disclosed in the methods document but missing from this report's own AI statement, a count retyped wrongly **inside the organizers' .xlsx methods form** — which the gate could not read at all until 2026-10-20, and which turned out to be carrying two stale counts and a withdrawn global negative the moment it could — a missing deliverable, and a stated suite size that disagrees with the number of tests on disk (the one count the gate computed nothing for, and the one that had drifted three times across four files). A final test runs the unperturbed tree and requires that none of those messages appears, because a checker that fails on everything proves nothing. The suite in `tests/phase1/` is 79 checks in total. Two fill slots — the frozen date and the pitch-video URL — are reported as PENDING before the freeze and become blocking under `make_manifest.py --freeze`, which is the submission-day run. The 28 self-tests pin the engine's behavior at its edges: unknown input blocks, a safety contradiction cannot reach STRONG, a clean input still can, the scenario gate rejects an invalid objective class, flags a conditional one, blocks an undeclared one, blocks a malformed one, and stays inert when no scenario is active — and five tests run against this patient's own constraint levels so that the gates' reject, require and flag branches are actually exercised. The engine demonstrates gate logic only; scientific judgment lives in the canonical table and this report, not in the code.

13. Limitations

Stated plainly, not defended. **Candidate generation was literature-driven, not a database sweep.** Entries were produced by three independent adversarial agent passes over the primary literature, cross-checked against a competitor's public repository, with the originating channel and any convergence recorded per candidate (`registers/candidate_register.csv`, which uses the earlier K/R/A/E identifiers and is kept as the historical channel record; the canonical candidate layer moved to `docs/candidates_canonical.json` on 2026-10-08) — several entries, including the whole wrong-direction mitotic-inhibitor group, were reached by all three channels independently. What this is *not* is a systematic screen over DGIdb / ChEMBL / Open Targets / connectivity-map data, so a non-obvious candidate that only such a screen would surface could have been missed; we do not claim that coverage. Agent passes are a search strategy, not evidence: every surviving claim is backed by a primary source, and agreement between agents is never counted as corroboration. Relatedly, the decision rule in §1 was written alongside the candidate work rather than pre-registered before it — unlike our Track 1 ranking rules, which were committed in advance with a timestamp. We state both rather than implying a rigour we did not have. The `objective_class` assignment that the scenario analysis turns on (§8) was likewise a single human pass, recorded as a judgment rather than a measurement. **One citation moved twice in a single day, and both moves were ours to own**: the SIRT2–K250 link (§3 link 7c, chain edge E16b) was first graded `asserted` because we could not obtain the source text, and was simultaneously written up as an opposing effect on BubR1 abundance — under-grading the evidence and over-stating the objection at the same time. On obtaining the abstract we upgraded the grade to `measured` and narrowed the claim to what the source actually supports. The correction favours our own hypothesis, which is the reason it is disclosed rather than silently applied. The automated citation check (`scripts/phase1/16_verify_citations.py`) resolves every PMID against Europe PMC and compares the returned title with the one we recorded. It originally read only the mechanism graph and the claim ledger, which between them miss about two thirds of the citations a reader can see — a PMID cited in the report's prose, in the safety annex or in the eligibility matrix is not a graph-edge source. It now reads all of those too: **36 load-bearing PMIDs, every one with a recorded title to compare against**. Of the 26 rows in `citation_check.csv` that the automated comparison leaves as `manual`, 22 are UniProt feature annotations pulled in by `--include-uniprot` and carry no part of the argument; the other four are papers cited in the report's prose and in the all-MVA table (PMIDs 41820377, 39490401, 41700350, 39845481), whose identifiers resolved but which carry no recorded title hint to compare against, because a citation in prose has nowhere to record one. Their retrieved titles are printed in the register and are checked by hand; this is a structural limit of the check, not an unresolved mismatch. No NAD+ booster has been tested in an MVA model or in cells of this allele class — the niacin step is a bridging hypothesis. NAD+ status in MVA has never been measured, so the deficiency the intervention would correct has not been shown to exist. The SIRT2–BubR1 effect is also background-specific: it does nothing on a normal background (Wu 2023), which is reassuring for specificity but means no general evidence supports it. **The therapeutic chain rests on two papers and we can say exactly which — and the graph as a whole rests on five.** Leave-one-source-out (§8b) shows that removing North 2014 leaves chain edges E15 and E16 unsourced and removing Pirinen 2020 leaves E14 unsourced; either cut severs every path from niacin to residual BubR1. One paper carries the whole SIRT2→BubR1 core. We previously wrote that the nearest independent work on that axis is Wu

2023, a *non*-replication at the organismal endpoint; that was not right, and the correction runs in our own favour, which is why it is here. The nearest independent work is **Qiu 2018** (PMID 29067790), from a different group, and it runs *with* the axis: depleting Sirt2 damages meiotic spindle and chromosome organisation, the acetylation-mimetic BubR1-K243Q reproduces that damage, and the non-acetylatable K243R — the state SIRT2 activity produces — partly prevents it and lowers the incidence of aneuploid eggs from aged mice. What it is not is a replication of the step this hypothesis needs: a different lysine, meiosis rather than mitosis, a mutant over-expressed rather than endogenous protein raised, and no drug anywhere in it. So the honest statement is narrower than either of our two previous ones: the K668 *abundance* mechanism has no independent replication, the acetylation axis as a whole has one independent corroboration at an aneuploidy endpoint, and the only independent test of the organismal endpoint (Wu 2023) is a non-replication on a normal background. The honest reading is that this is a single-source mechanism dressed in a multi-citation chain, and a reader who doubts North 2014 should set the hypothesis aside rather than discount it. Residual BubR1 function in this child is unmeasured; the candidate holds only under scenario S1. There is no pediatric niacin label. The oncology direction is unresolved. Phase is assumed, not measured. Pathogenicity predictors are predicted, not measured, and are not used as if measured. The engine's gate verdicts are not scientific disposition. Systemic NAD+ repletion is not uniform across tissues — in the adult human study the rise in blood was substantially larger than the rise in muscle — so a blood NAD+ response would not by itself establish that the relevant tissues reached a BubR1-stabilizing exposure. The ciliary branch is supported at the model-organism level only and its readouts are marked exploratory here, not claimed. The decisive experiment requires patient-derived cells this team does not access; we specify it, we do not perform it.

14. Methods and AI statement

Full detail in `METHODS_AND_AI.md`. In brief: mechanism and therapeutic objective were frozen before candidate reasoning; candidates passed directionality and four patient gates; each lead met an adversarial review instructed to refute it; every load-bearing citation passed an automated title-resolution check (36 PMIDs, no unresolved mismatches) and an independent human verification pass, both re-run end-to-end on 2026-10-19 after a second adversarial review and recorded in `registers/citation_manual_check_2026-10-19b.md` with its own access limits stated. That round removed one miscited source from the mechanism chain, downgraded one edge from measured to asserted because the source's human result is a prediction, and rejected one claim the reviewer had attributed to a surveillance consensus that does not make it. AI usage: Claude Max (claude.ai and Claude Code) with model-training opt-out enabled in account settings and no feedback given on any output, under Anthropic Consumer Terms, **and a second provider (OpenAI, ChatGPT/Codex) used for independent adversarial review under the same no-feedback, training-off handling**; the full provider list with per-session handling is `registers/ai_usage_register.csv`, and the release manifest blocks if a provider appears in this report, in `METHODS_AND_AI.md` or in that register without appearing in the other two; inputs to AI were routinely limited to derived information (HPO terms, protein-level variant representations, mechanism text, code, public literature), and no raw data file was ever opened by an AI tool — every step that reads patient data ran as a plain script in the submitter's own terminal. We additionally disclose two occasions on which coordinate-level content nevertheless reached an assistant conversation: a post-run terminal paste in Track 1, and a working-ledger row seen during an internal audit in Track 2. Our Track 1 methods document states the same thing in the same terms, so a panel comparing the two will find them consistent; both are permitted under the organizers' final AI guidance. **Two** responses were interrupted by the provider's safety classifier — one in a literature-only session (2026-10-14..18) and one on 2026-10-19 while summarising tool output. Neither session contained genotype-level data and no feedback was given on any output. The count here is kept identical to `METHODS_AND_AI.md` and `registers/ai_usage_register.csv`; an earlier draft of this sentence said "one", which was an undercount, and the release manifest now cross-checks the number between this report and `METHODS_AND_AI.md` so it cannot drift apart again; the register itself is the human-maintained record behind both. Settings evidence is archived and available on request.

15. Acknowledgement and citation

> "*This work was made possible through the Hackathon, organized by Sage Bionetworks in partnership with the MVA Society, Hugging Face, and BEACON (The Benchmarking, Evaluation, and Assessment Consortium for Science), with prize sponsorship from AWS and Anthropic. We are deeply grateful to the child and their family who generously contributed their data and their story to advance research into this rare disease. We acknowledge their trust in making this Hackathon possible.*"

The Hackathon dataset is cited using the reference provided on the Hackathon Synapse page at the time of publication. **What this submission actually contains, stated precisely.** The report body describes the phenotype at category level only. The structured files that the engine reads — `config/cases/bub1b_mva.yaml` and `docs/mechanism_graph.yaml` — carry the eight phenotype observations as named HPO terms with their HP identifiers, because the mechanism chain has to connect specific terms and because the organizers' guidance lists HPO terms among the derived, shareable facts. Variant representation is at transcript and protein level (c. / p. HGVS on a MANE transcript); **no genomic coordinate, no genotype field, no sample identifier and no raw data file appears anywhere in this repository**, and `scripts/phase1/19_public_safety.py` enforces that on every commit — it now also warns on any `HP:` identifier so that each one is a deliberate choice rather than an oversight. **That guard had a hole, and it is worth recording how it was found.** On 2026-10-20 a red-team pass wrote coordinates into a submission document in the three forms a coordinate actually arrives in — a position with its alleles attached, as a browser or a VCF prints it; the same thing with dashes or underscores throughout, as an annotation tool prints it; and a position carrying a deletion suffix — and the scan reported no coordinates found and exited clean, because its pattern required the position to be followed by a non-word character, and the allele letter that makes the string a variant is a word character. A guarantee enforced by a checker is worth exactly what the checker catches, so the forms are now matched, the pre-commit hook blocks them, and the case is pinned by a test that fails against the pre-fix pattern.

We considered reducing the structured files to category labels as well. We did not, because the mechanism graph's value is that each phenotype term is connected to a sourced edge or explicitly left unexplained, and category labels would make that check unverifiable. We note the residual consideration plainly: in a disease with a few hundred reported cases worldwide, a biallelic *BUB1B* genotype combined with eight specific terms is a narrow description, and anyone reusing this repository should weigh that before adding further detail.

Code and this report: GitHub `https://github.com/Vero79L/mva-hackathon-2026-track2` *(private during the Hackathon; public at close under CC BY 4.0). Pitch video:* `{{FILL:VIDEO}}`*. Frozen* `{{FILL:DATE}}`*.*